WORKED SOLUTION

STEP II 2016, Question 11

Mechanics • Projectile and target

STEP II 2016, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/16-s2-q11

The key idea

Two particles collide when both coordinates agree at the *same* time, so we get two equations sharing one unknown t; dividing one by the other removes t. In part (ii) the trick is to notice that gravity pulls the bullet and the target down by exactly the same amount, $\frac{1}{2}gt^2$, at every instant. So whether they meet is decided by the gravity-free motion of part (i), and gravity only matters for asking where the target is when they meet.

Solution

Part (i)

Call the first particle P (starting at $(a,0)$) and the second Q (starting at $(0,b)$). Suppose they collide at time t. Equating the x- and y-coordinates,

$$\begin{aligned} a + ut\cos\alpha &= vt\cos\beta, && \text{so} \quad t\,(v\cos\beta - u\cos\alpha) = a,\\ ut\sin\alpha &= b + vt\sin\beta, && \text{so} \quad t\,(u\sin\alpha - v\sin\beta) = b. \end{aligned}$$

Since $a > 0$, the first equation shows $t \neq 0$ (the particles start at different points), so $t > 0$ and we may divide the second equation by the first:

$$\frac{u\sin\alpha - v\sin\beta}{v\cos\beta - u\cos\alpha} = \frac{b}{a} = \tan\theta = \frac{\sin\theta}{\cos\theta}.$$

(The denominator is $a/t \neq 0$.) Cross-multiplying,

$$\cos\theta\,(u\sin\alpha - v\sin\beta) = \sin\theta\,(v\cos\beta - u\cos\alpha),$$

and collecting the u terms on the left and the v terms on the right,

$$u(\sin\alpha\cos\theta + \cos\alpha\sin\theta) = v(\sin\beta\cos\theta + \cos\beta\sin\theta).$$

By the compound-angle formula for $\sin(A+B)$ this is

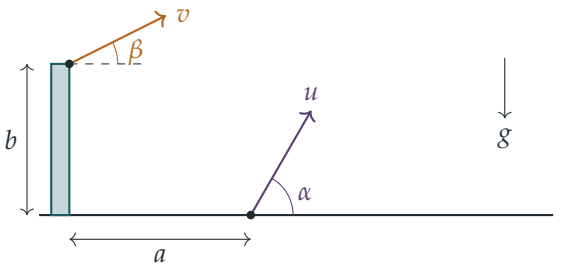
$$u\sin(\theta+\alpha) = v\sin(\theta+\beta),$$

as required. We also record two facts for later: since $a, b, t > 0$,

$$v\cos\beta > u\cos\alpha \qquad \text{and} \qquad u\sin\alpha > v\sin\beta. \tag{$*$}$$

Part (ii)

Model the bullet and the target as particles moving freely under gravity, with no air resistance. Take the foot of the tower as the origin, the x-axis horizontal through the target's launch point (in the direction away from the tower), and the y-axis vertically up. The bullet must be fired towards the target for a hit to be possible, so both move in this vertical plane.



At time t (before either lands), the target and bullet are at

$$T = (a + ut\cos\alpha,\ ut\sin\alpha - \tfrac{1}{2}gt^2), \qquad B = (vt\cos\beta,\ b + vt\sin\beta - \tfrac{1}{2}gt^2).$$

These are exactly the positions of part (i), each shifted down by the same amount $\frac{1}{2}gt^2$.

> **Ask yourself** Why can we use part (i) when part (i) had no gravity?

The hit happens when $T = B$. In the y-equation the term $-\frac{1}{2}gt^2$ appears on both sides and cancels, so the collision conditions are word for word those of part (i):

$$t\,(v\cos\beta - u\cos\alpha) = a, \qquad t\,(u\sin\alpha - v\sin\beta) = b.$$

In particular part (i) and $(*)$ apply, and the time of the hit is

$$t = \frac{b}{u\sin\alpha - v\sin\beta},$$

where the denominator is positive by $(*)$.

The target is still in the air at this moment exactly when its height is positive:

$$ut\sin\alpha - \tfrac{1}{2}gt^2 > 0 \iff t < \frac{2u\sin\alpha}{g},$$

dividing by $\frac{1}{2}t > 0$. Substituting the time of the hit,

$$\frac{b}{u\sin\alpha - v\sin\beta} < \frac{2u\sin\alpha}{g}.$$

Both denominators are positive, so multiplying through by $g(u\sin\alpha - v\sin\beta)$ keeps the direction of the inequality, and we obtain

$$2u\sin\alpha\,(u\sin\alpha - v\sin\beta) > bg,$$

as required. (At the hit the bullet is at the same height as the target, so it too is above the ground.)

Why we need $\alpha > \beta$

We have seen that a hit requires the two particles to collide in the sense of part (i), so

$$u\sin(\theta+\alpha) = v\sin(\theta+\beta),$$

where $\tan\theta = b/a$, and also, from $(*)$, $u\sin\alpha > v\sin\beta$. All four sines here are positive, because θ, α, β are acute, so $\theta+\alpha$ and $\theta+\beta$ lie strictly between 0 and π. Dividing the equation by the inequality reverses it:

$$\frac{\sin(\theta+\alpha)}{\sin\alpha} < \frac{\sin(\theta+\beta)}{\sin\beta}.$$

Expanding, $\dfrac{\sin(\theta+\alpha)}{\sin\alpha} = \sin\theta\cot\alpha + \cos\theta$, and similarly for β, so

$$\sin\theta\cot\alpha < \sin\theta\cot\beta \iff \cot\alpha < \cot\beta,$$

since $\sin\theta > 0$. On $(0, \frac{\pi}{2})$ the cotangent is strictly decreasing, so $\alpha > \beta$.

Physically: relative to the bullet, the target moves in a straight line at constant velocity, because gravity has cancelled. That relative velocity must point from the target's start $(a, 0)$ to the gun at $(0, b)$, so it must be backwards and upwards; by $(*)$ the target must be slower horizontally but faster vertically than the bullet, which forces its path to be steeper, $\tan\alpha > \tan\beta$. (Dividing the two parts of $(*)$ gives the same conclusion directly.)

Checks

Try $u = 20$, $\alpha = 60^\circ$, $v = 15$, $\beta = 30^\circ$ and a hit at $t = 1$, with $g = 9.8$. The collision equations give $a \approx 2.990$ and $b \approx 9.821$, so $\theta \approx 73.06^\circ$. Then $u\sin(\theta+\alpha) \approx 14.612$ and $v\sin(\theta+\beta) \approx 14.612$, agreeing. At the hit the target is at height $17.32 - 4.9 \approx 12.42 > 0$, and indeed $2u\sin\alpha(u\sin\alpha - v\sin\beta) \approx 340.2 > bg \approx 96.2$. A computer check over many thousands of random launches confirmed that the inequality holds exactly when the target is still airborne at the hit, and that every hit had $\alpha > \beta$.

Common slips

- Forgetting that the two particles must be at the same place at the *same* time, and solving for two different times.
- Multiplying through by $u\sin\alpha - v\sin\beta$ without first establishing that it is positive; it is, because $t > 0$ and $b > 0$.

- Arguing $\alpha > \beta$ from $u \sin\alpha > v \sin\beta$ alone, which says nothing about the angles unless the speeds are compared too.

Beyond STEP

The cancellation of gravity is a case of Galilean relativity: in a frame falling freely with the bullet, the target moves in a straight line at constant velocity, so "aim straight at it" works. This is the classic "monkey and hunter" demonstration, and the idea of an inertial (or freely falling) frame is the starting point of first-year mechanics and, later, of the equivalence principle in general relativity. Kleppner and Kolenkow, *An Introduction to Mechanics*, develops relative motion and reference frames carefully.

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